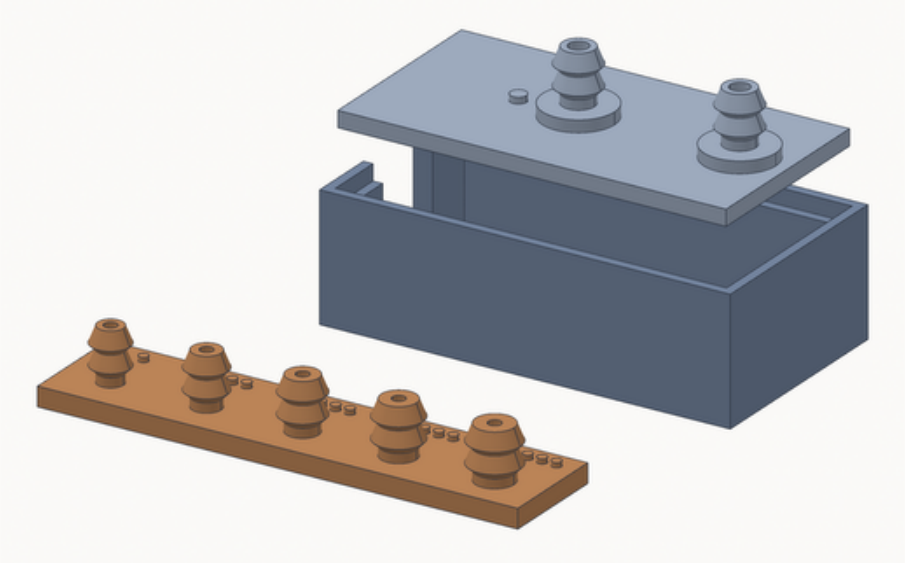
Team 6 · 3D print request · version 1

EXSU 620 Project 06, exhaled CO2 monitor: sensor flow-cell and tubing test piece · STEP files · 22 Oct 2026



1 Tubing-fit coupon (orange)

2 Tray (blue)

3 Lid (light blue)

Order	File	Size (mm)	Place on the bed	Supports	Rough time
1 (first)	barb_fit_coupon_v1.step	60 × 14 × 9	flat side down	none	~10 min
2	flowcell_tray_v1.step	51.3 × 27.5 × 15.6	open side up	none	~45 min
3	flowcell_lid_v1.step	48.5 × 24.7 × 10.1	flat side down, barbs up	none	~20 min

Settings

- PLA, 0.4 mm nozzle, 0.2 mm layers (0.12 to 0.16 mm is nicer for the barbs if you have time)
- 3 to 4 walls, 20 to 30 % infill; no brim needed
- Minimum layer time about 10 s, so the small barbs cool between layers
- Every part is designed to print without supports

Notes

- STEP files (AP214) in millimetres; the same parts are also available as STL.
- Version 1 is sized from the sensor datasheet, before we measured; small changes may follow.
- Please print the coupon first if you can: it tells us which barb size grips our oxygen tubing.
- The tray and lid hold the CO2 sensor; the lid drops into a ledge and the seam is taped.
- Times are rough; your slicer will give the real ones. About 10 to 15 g of PLA in total.

Could you tell us

- Which printer and nozzle you use, and how much holes usually come out undersize
- Whether you can print flexible TPU (for a gasket); otherwise we use foam tape
- Your usual turnaround time

Files: hardware/cad/ in the Team 6 project folder. make_parts.py (STL) and make_step.py (STEP) rebuild every part from the measurements.